

[Soln:] Suppose A is a Knight. $\therefore$ A always speaks the truth, then the statement "Atleast one of us is a Knave" infers that B is a Knave. Suppose A is a Knave. $\therefore$ "Atleast one of us is a Knave" is True which is a contradiction to the fact that a Knave always lies. $\therefore$ A cannot be a Knave.

$A \rightarrow \text{Knight} \quad B \rightarrow \text{Knave}$

24) A says "The two of us are both Knights" and B says "A is a Knave".

[Soln:] Suppose A is a Knight. $\therefore$ A Knight always speaks the truth and inferring from A's statement, we conclude B is also a Knight. B says "A is a Knave". Since B is a Knight, we can conclude that A is a Knave. This is a contradiction to the fact that A is a Knight.

$\therefore$ A is a Knave. $\therefore$ a Knave always lies, the two of them are not both Knights.

B's statement that A is a Knave is true. $\therefore$ B is a Knight

$A \rightarrow \text{Knave} \quad B \rightarrow \text{Knight}$

25) A says "I am a Knave or B is a Knight" and B says nothing.

[Soln:] Suppose A is a Knight. $\therefore$ The statement "I am a Knave or B is a Knight" is True. By defn. of OR, the following 3 cases arises.

[Case 1:] Only I am a Knave is True. We know, since A is a Knight, A is not a Knave, and "I am a Knave" is False. So, contradiction

[Case 2:] Both "I am a Knave" and "B is a Knight" are True. By case 1, this can never happen.

[Case 3:] Only "B is a Knight" is True. $\therefore$ A $\rightarrow$ Knight B $\rightarrow$ Knight

Suppose A is a Knave. $\therefore$ The statement "I am a Knave or B is a Knight" is False. By defn. of OR, "I am a Knave" is False and "B is a Knight" is False. $\therefore$ This is a contradiction since A is a Knave. $\therefore$ A cannot be a Knave